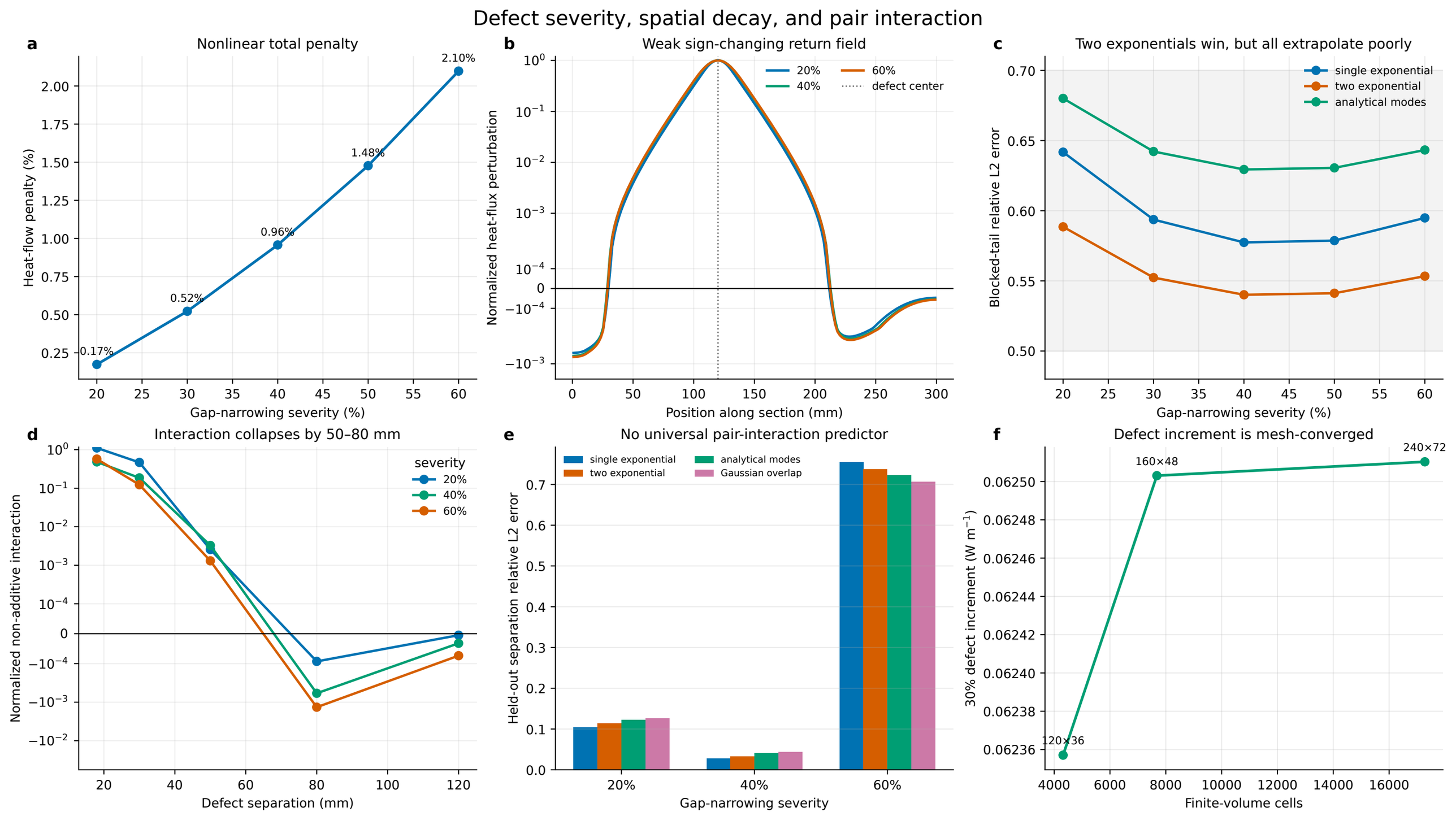
Defect severity, spatial decay, and pair interaction
a
Nonlinear total penalty
Heat-flow penalty (%)
Gap-narrowing severity (%)
0.17%
0.52%
0.96%
1.48%
2.10%
b
Weak sign-changing return field
Normalized heat-flux perturbation
Position along section (mm)
20%
40%
60%
defect center
c
Two exponentials win, but all extrapolate poorly
Blocked-tail relative L2 error
Gap-narrowing severity (%)
single exponential
two exponential
analytical modes
d
Interaction collapses by 50–80 mm
Normalized non-additive interaction
Defect separation (mm)
severity
20%
40%
60%
e
No universal pair-interaction predictor
Held-out separation relative L2 error
Gap-narrowing severity
single exponential
two exponential
analytical modes
Gaussian overlap
f
Defect increment is mesh-converged
30% defect increment (W m^{-1})
Finite-volume cells
120×36
160×48
240×72